

CHAPTER 6

Noah Parker

6.1. (a) Let M be a Noetherian A -module and $u : M \rightarrow M$ a module homomorphism.

If u is surjective, then it is an isomorphism.

(b) If M is Artinian and u is injective, then again u is an isomorphism.

Proof. (a) Consider the filtration $\ker u \subset \ker u^2 \subset \cdots$, and the associated complex of modules

$$\cdots \rightarrow \frac{\ker u^2}{\ker u} \xrightarrow{u} \ker u \rightarrow 0.$$

Note that $u : M \rightarrow M$ is surjective $u^{-1}(\ker u^k) = \ker u^{k+1}$, so each of the above maps is surjective. It follows by induction that the complex is zero if and only if it is bounded. M Noetherian gives us $n > 0$ such that $\ker u^n = \ker u^{n+1}$, whence the complex is bounded, and so $\ker u = 0$ as desired.

(b) Consider the filtration $\cdots \subset \operatorname{im} u^2 \subset \operatorname{im} u$, and the associated complex of modules

$$0 \rightarrow \operatorname{cok} u \xrightarrow{u} \frac{\operatorname{im} u}{\operatorname{im} u^2} \rightarrow \cdots$$

The kernel of each map is of the form $u^{-1}(\operatorname{im} u^{k+1}) = u^{-1}(u(\operatorname{im} u^k)) = \operatorname{im} u^k$ by injectivity of u , so each map is injective. It follows by induction that the complex is zero if and only if it is bounded. M Artinian gives us $n > 0$ such that $\operatorname{im} u^n = \operatorname{im} u^{n+1}$, whence the complex is bounded, and so $\operatorname{cok} u = 0$ as desired. $\square$

6.2 Let M be an A -module. If every non-empty set of finitely generated submodules of M has a maximal element, then M is Noetherian.

Proof. Let Σ be a non-empty set of submodules of M , and let Σ' be the set of finitely generated submodules of elements of Σ . Then there is a maximal element $N' \in \Sigma'$, which is in particular finitely generated and contained in some $N \in \Sigma$. It suffices to show that N is a maximal element of Σ .

Suppose that $N \subsetneq P \in \Sigma$, and take some $x \in P \setminus N$. Then $N' \subsetneq (N', x) \subset P$ is contained in Σ' , contradicting maximality of N' . Hence, $N \in \Sigma$ is maximal, whence M is Noetherian. $\square$

6.4. Let M be a Noetherian A -module, and let $\mathfrak{a}$ be the annihilator of M in A . Prove that $A/\mathfrak{a}$ is a Noetherian ring.

If we replace *Noetherian* by *Artinian*, is this result still true?

Proof. M Noetherian gives us generators $x_1, \dots, x_n$ of M , so that $\mathfrak{a} = \bigcap_{i=1}^n \operatorname{Ann}(x_i)$. Further, $M^{\oplus n}$ is Noetherian by 6.46 Then we have an A -module morphism

$$\begin{aligned} \phi : A &\rightarrow M^{\oplus n} \\ r &\mapsto (r \cdot x_i)_{i=1}^n \end{aligned}$$

with $\ker \phi = \mathfrak{a}$, so that $A/\mathfrak{a} \cong \operatorname{im} \phi \leq M^{\oplus n}$ is a Noetherian A -module. By (6.6), $A/\mathfrak{a}$ is a Noetherian ring, as desired.

The result does not hold if we replace Noetherian by Artinian. Consider the case of $A = \mathbb{Z}$ and $M = G$ the subgroup of $\mathbb{Q}/\mathbb{Z}$ consisting of elements whose order is a power of p , where p is a fixed prime. Then G is Artinian, as it has only the subgroups $G_0 \subset G_1 \subset \cdots \subset G_n \subset \cdots$ of order p^n for each $n \geq 0$. However $\mathfrak{a} = \bigcap_{n \geq 0} p^n \mathbb{Z} = (0)$, so that $A/\mathfrak{a} \cong \mathbb{Z}/(0) \cong \mathbb{Z}$ is not Artinian by considering the chain (p^n) for $n \geq 0$. $\square$

6.11. Let $f : A \rightarrow B$ be a ring homomorphism and suppose that $\text{Spec } B$ is a noetherian space. Prove that $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is a closed mapping if and only if f has the going up property (5, Ex. 10).

Proof. Suppose that f has the going up property. Take a radical ideal $\mathfrak{b} \trianglelefteq B$. Then

$$f^*(V(\mathfrak{b}))$$

$\square$

6.12 Let A be a ring such that $\text{Spec } A$ is Noetherian. Show the set of prime ideals of A satisfies the ascending chain condition. Is the converse true?

Proof. Let $\mathfrak{p}_1 \subset \mathfrak{p}_2 \subset \cdots$ be an ascending chain of prime ideals $\mathfrak{p}_n \trianglelefteq A$. Then we get a descending chain of closed sets

$$V(\mathfrak{p}_1) \supset V(\mathfrak{p}_2) \supset \cdots$$

in $\text{Spec } A$, which stabilizes by $\text{Spec } A$ Noetherian. Thus, there is $n \geq 1$ such that $V(\mathfrak{p}_n) = V(\mathfrak{p}_{n'})$ for all $n' \geq n$. Prime implies radical, so in fact $\mathfrak{p}_n = \mathfrak{p}_{n'}$ for all $n' \geq n$. We conclude that the set of prime ideals of A satisfy the ascending chain condition, as desired.

Need to show the converse doesn't hold. Must find a non-Noetherian ring with non-Noetherian spectrum such that the prime ideals satisfy ACC. Can't use $k[x_0, \dots, x_n]$, a

Consider the ring $C(X)$ of real-valued continuous functions on a compact infinite Hausdorff space X . Probably this? $\square$